23-09-2026

Standard 12

BIOLOGY

Time Allowed: 3.00 Hours Maximum Marks: 70

BIO-BOTANY

Marks : 35

I. Choose the best answer: 8×1=8

1) Transmitting tissue is found in _____.
   a) Micropylar region of ovule b) Pollen tube wall
   c) Stylar region of gynoecium d) Integument
2) Coelorhiza is found in _____.
   a) Paddy b) Bean c) Pea d) Tridax
3) Which one of the following is an example of polygenic inheritance?
   a) Flower colour in Mirabilis Jalapa b) Production of male honey bee
   c) Pod shape in garden pea d) Skin colour in humans
4) In his classic experiments on pea plants mendel did not use _____.
   a) Flowering position b) Seed colour
   c) Pod length d) Seed shape
5) An allohexa ploidy contains _____.
   a) six different genomes
   b) six copies of three different genomes
   c) two copies of three different genomes
   d) six copies of one genome
6) The number of chromosomes in ophioglossum is _____.
   a) 216 b) 1262 c) 34 d) 24
7) ECORI cleaves DNA at _____.
   a) AGGGTT b) GTATATC c) GAATTC d) TATAGC
8) Solidifying agent used in plant tissue culture _____.
   a) Nicotinic acid b) Cobaltous chloride
   c) EDTA d) Agar

II. Answer any four questions: 4×2=8

9) Give the examples for micropropagation performed plants.
10) What do you know about the word pBR332?
11) What is gene mapping?
12) Name the scientist to rediscovered Mendelism.
13) What is meant by hybrid vigour?
14) What is Mellitophily?

III. Answer any three questions: Question No.19 is compulsory. 3×3=9

15) What is polyembrynony? How it can commercially exploited?
16) Differenciate incomplete dominance and codominance.
17) Write the salient features of Sutton and Boveri concept.
18) Name the chemical used in gene transfer.
19) Note on Cybrid.

IV. Answer all questions: 2×5=10

20) Explain the application of plant tissue culture.

(OR)

How is Nicotiana exhibit self incompatibility. Explain its mechanism.

21) Bring out the inheritance of chloroplast gene with example.

(OR)

Draw and label the Study Materials.

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I. Choose the best answer: **8×1=8**

1) Thelytoky is observed in _____.
   a) Solenobia b) Aphis gossypii
   c) Apis d) Mosquitos
2) Colostrum is also rich in _____ antibodies.
   a) IgA b) IgE c) IgG d) IgD
3) Non-medicated IUD is _____.
   a) Nova T b) CU 7 c) LNG 20 d) Lippes loop
4) The number of functional gene in 'Y' chromosome is _____.
   a) 1000 b) 1200 c) 60 d) 600
5) A mRNA molecule is produced by _____.
   a) Replication b) Transcription c) Duplication d) Translation
6) Which period was called "Age of fishes"?
   a) Permian b) Triassic c) Devonian d) Ordovician
7) Cirrhosis of liver of caused by chronic in take of _____.
   a) Opium b) Alcohol c) Tobacco d) Cocaine
8) Allergy involves _____.
   a) IgA b) IgG c) IgM d) IgE

II. Answer any four questions: **4×2=8**

9) What are interferons? Mention their role.
10) What ae Coacervates?
11) What is criss cross inheritance?
12) How is STD transmitted?
13) What is the composition of semen?
14) What is meant by Apolysis?

III. Answer any three questions: Question No.19 is compulsory. **3×3=9**

15) How is Jevenile phase different from reproductive phase?
16) What is meant by 'LH Surge'?
17) Explain (a) ZIFT (b) ICSI.
18) What is Lyonisation?
19) Draw and label the diagram of Immunoglobulin.

IV. Answer all questions: **2×5=10**

20) What is Karyotyping? How it is prepared? Mention its application.

(OR)

Objections of Darwinism.

21) Explain the role of Oxytocin and Relaxin in parturition and lactation.

(OR)

Explain infertility in brief.

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